Aman Vishwakarma

CS Undergraduate | Software Developer

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Education

- **Indian Institute of Information Technology, Lucknow** July 2023 - June 2027
  B.Tech in Information Technology - CGPA: 8.083

Experience

Google Summer of Code 2025 & 2026 Contributor @ Zulip May 2025 - Sept 2025, May 2026 - Aug 2026

- **Contributions (PRs)** : Wrote 5,000+ lines of industry-level code. Contributed 60+ commits across 45+ PRs in backend and frontend areas.
- **Mark messages as read modal (new feature)**: Integrated with Zulip's `bulk_update_read_flags_for_narrow` API to support three message-type scopes; handled server-side 50K unread cap edge case and wrote unit tests.
- **Mark a user as spammer (new feature)**: Extended Zulip's user-deactivation Django API with an `is_spammer` flag triggering cascaded profile/message deletion across user-owned bots; optimized `do_delete_messages` to accept pre-fetched `streams_by_recipient_id` dict, eliminating redundant per-stream DB queries.
- **Refactor API to group messages by topic (API refactor)**: Refactored `do_delete_messages` to group messages by recipient-topic using a `defaultdict`, batching delete events per unique topic to comply with Zulip's API contract of one event per topic.

Shinpo Engineering PTE LTD | Full-Stack Developer Intern Oct 2025 – Mar 2026, Singapore (Remote)

- Integrated Strapi uploads with **AWS S3** via env-driven bucket/region config, standardizing large file handling up to 250MB per upload with matching request/body caps for production use.
- Hardened Strapi's security middleware by configuring a **custom CSP** that allowlists **S3 bucket origins** for `img-src`/`media-src` while preserving framework defaults; enforced request and multipart size limits aligned to upload thresholds, reducing attack surface for malicious content injection.
- Productionized **EC2** runtime with **pm2 + nginx**; delivered 2 cron runners with flock-based lockfiles preventing overlapping executions, structured logging, and an event-driven listener option for async job triggering; configured **CloudWatch alarms** for EC2 instance monitoring

Projects

- **SecureGate | Go, Redis, Prometheus, Grafana, Docker Compose**
  - Built a security-first Go API gateway that enforces **JWT authentication + RBAC authorization** at the edge to prevent unauthorized access/privilege escalation.
  - Implemented **Redis-backed rate limiting** (INCR/EXPIRE) to block abusive traffic, with **Prometheus** counters tracking **rate-limit breaches for operational visibility**.
  - Shipped end-to-end **observability**: exported **Prometheus metrics** (request volume + latency histograms + auth/rate-limit failure signals) and wired **Prometheus + Grafana** via **Docker Compose for monitoring/alerting workflows**.

Achievements

- **Google Summer of Code'25 and 26 @ Zulip**: Contributed **60+ commits** over **45+ PRs**.
- **Leetcode**: **Knight (1933)** on Leetcode.
- **Codeforces**: Maximum rating **1450 (Specialist)** on Codeforces.
- **Codechef**: Maximum rating **1677 (3-star)** on Codechef.
- Solved 1,000+ DSA and competitive programming problems across various platforms.

Skills

- **Backend**: Go/Golang, System Design, REST APIs, CI/CD, Node.js, Express.js, Microservices
- **Security & Observability**: Prometheus, Grafana, JWT, RBAC, Rate Limiting, Alerting, Anomaly Detection
- **Cloud & Infrastructure**: AWS, Docker, Git, Linux, Distributed Systems
- **Languages**: JavaScript, Typescript, Go/Golang, C++, C, Python, SQL, Bash
- **Databases**: PostgreSQL, MySQL, Redis, MongoDB
- **Academic Coursework**: Computer Networks, Object Oriented Programming, Operating System, DBMS, DSA, Software Engineering

Positions of Responsibility

- **FOSS Wing Lead - Google Developers Group, IIIT Lucknow** Aug 2025 - Present
  Conducting Git tutorials and delivering technical presentations on open-source for over 200 students.